



SNAPCASH

GROUP 3:

Christofer Marveleous Krisdianto - 2702260331

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Kennes Jansen - 2802484804

Angelo - 2802480421

Meini Rusiadi - 2802467016

SNAPCASH



SNAPCASH

Create a better future for SMEs.



\$NAPC.ASH
QUICK, EASY, SNAP.

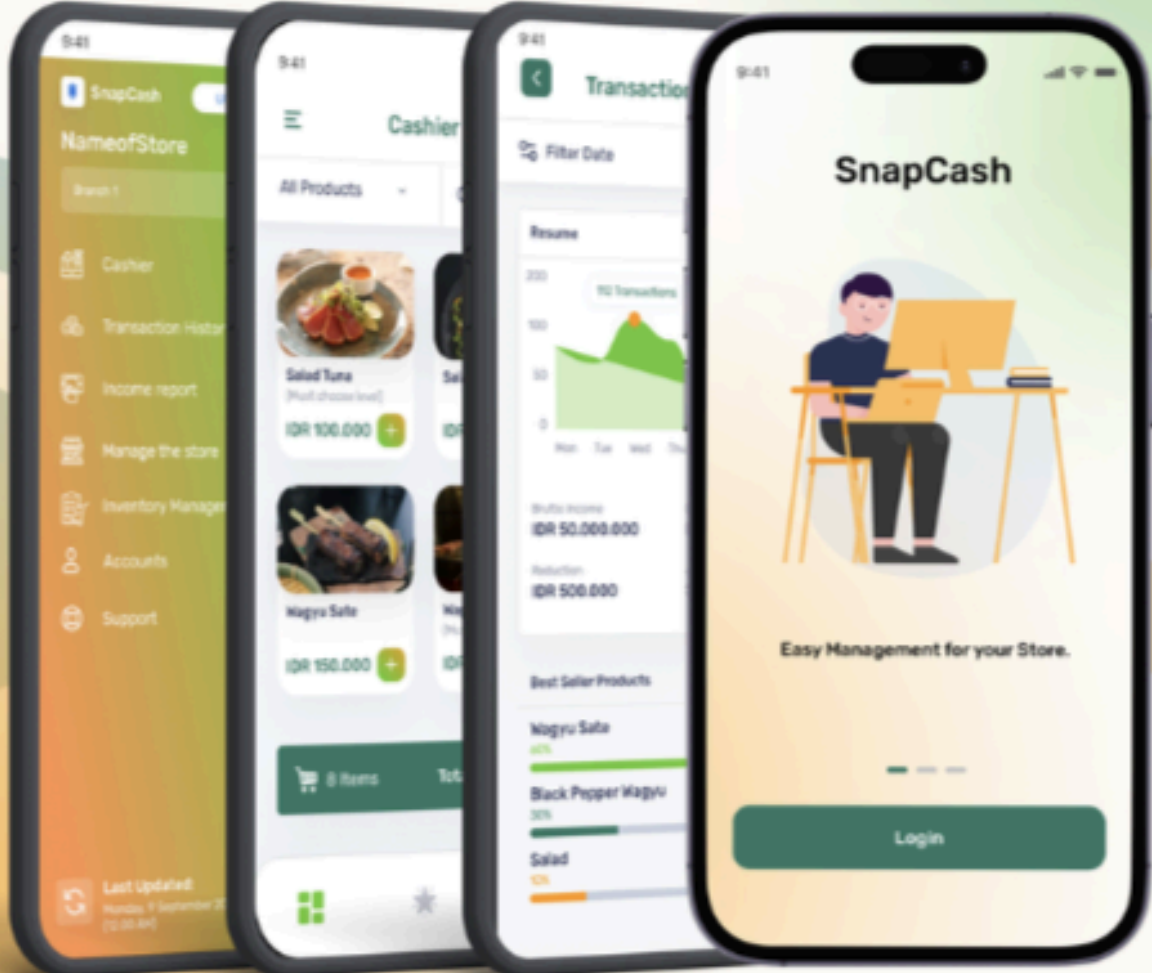
WA : 0814-1224-8658 (Marvel)

WA : 0811-2387-300 (Tracy)

WA : 0857-5452-7330 (Cecy)



Aplikasi Kasir untuk para Entrepreneur



Yuk Coba Fitur kami! Bisa Scan Disini →



SNAPCASH

\$NAPC.ASH

Fitur Inventory Management di Aplikasi Kasir untuk para Entrepreneur

Fitur Owner & Karyawan

Kasir Online

Inventory

Akutansi

Supply Chain

Analisa Bisnis

Yuk Coba Fitur kami! Bisa Scan Disini →

Pricing Options

Standard Package

Perfect for growing businesses seeking for efficient cashier management

Rp 100.000/month

- Cashier systems
- Inventory management systems
- Employee & owner features

Premium Package

Designed for established businesses aiming for comprehensive POS management.

Rp 300.000/month

- Cashier systems
- Inventory management systems
- Employee & owner features
- Supply Chain management
- Customized POS management Strategy

Enterprise Package

Tailored for large businesses requiring extensive business management and dedicated support.

Rp 1.500.000/month

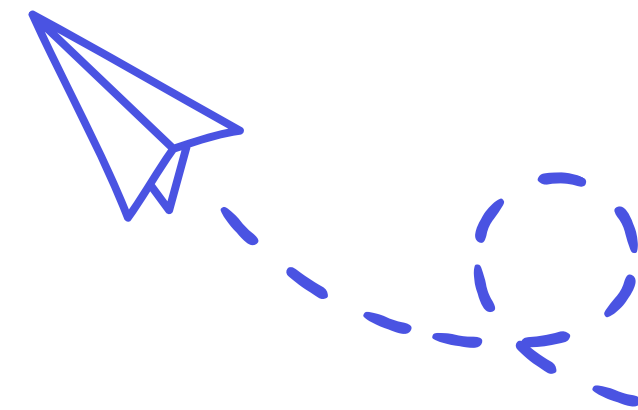
- Cashier systems
- Inventory management systems
- Employee & owner features
- Supply Chain management
- Customized POS management Strategy
- Branch per branch integrated management POS system

*price are outside of tax and hardware price

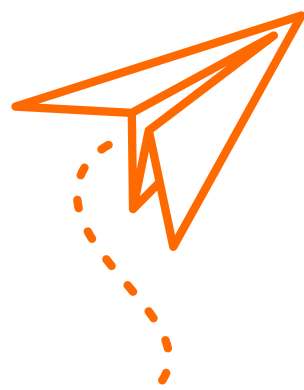


BACKGROUND

- MSMEs are a critical pillar of Indonesia's economy, yet many still rely on manual systems
- The FnB sector faces recurring challenges in managing sales, inventory, and financial reporting
- Lack of an integrated digital solution leads to inefficiencies and poor business visibility
- SnapCash was founded after a team member witnessed these struggles firsthand in a family-owned FnB business



COMPANY DESCRIPTION



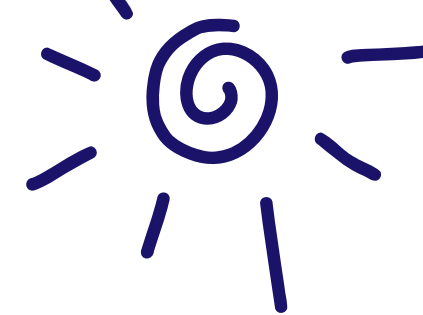


COMPANY DESCRIPTION

SnapCash is a digital Point of Sale (POS) platform designed to help Indonesian MSMEs, especially in the Food and Beverage (FnB) sector, manage their business operations more efficiently. The platform integrates transaction processing, inventory management, and financial reporting into a single easy-to-use system.

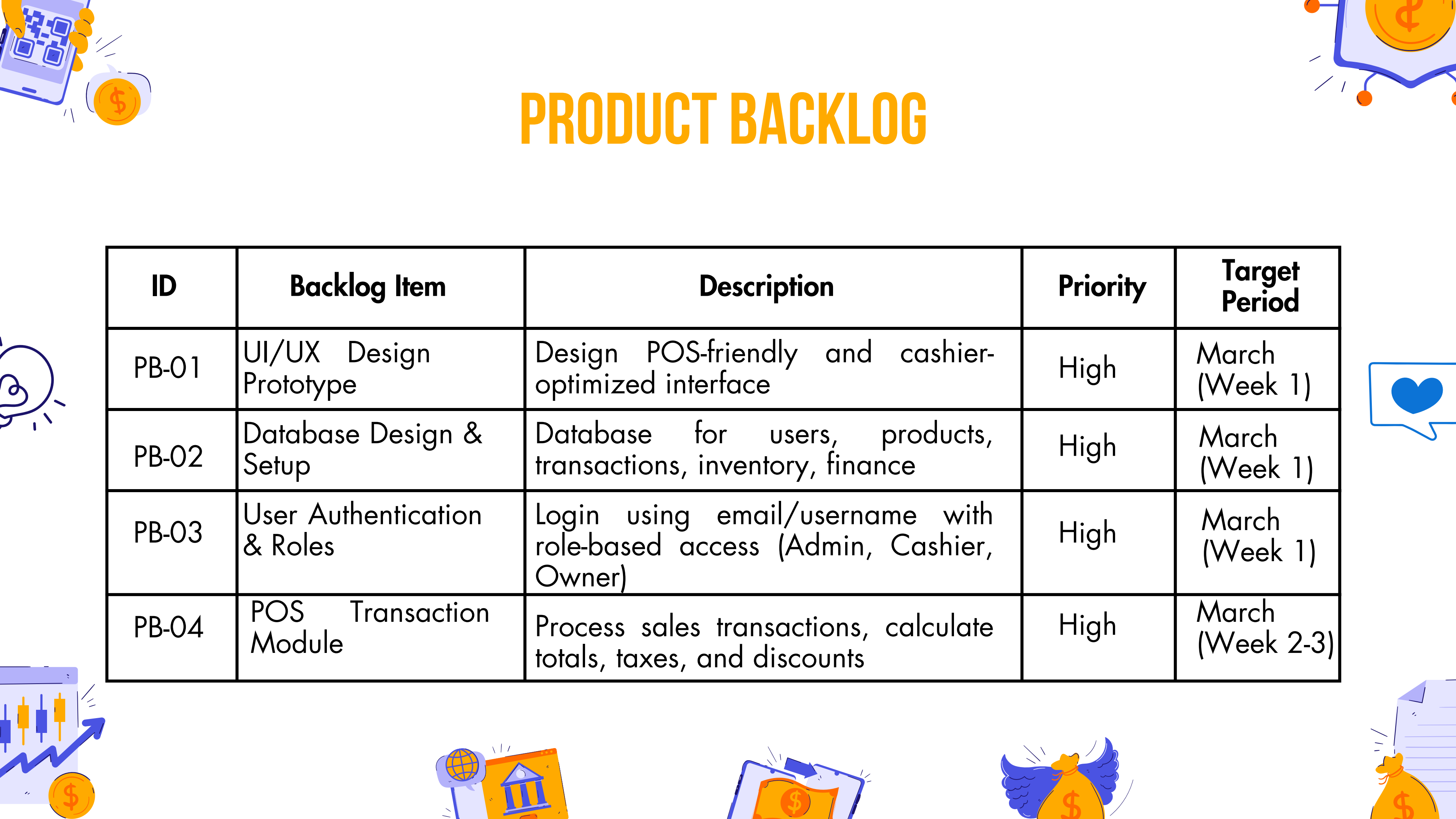
Developed by a team of Binus University students, SnapCash was inspired by real challenges faced by small businesses that still rely on manual operational processes.

With the tagline "Quick, Easy, Snap!", SnapCash aims to empower MSMEs through fast transactions, simple user experience, and data-driven business insights.



PRODUCT BACKLOG





PRODUCT BACKLOG

ID	Backlog Item	Description	Priority	Target Period
PB-01	UI/UX Design Prototype	Design POS-friendly and cashier-optimized interface	High	March (Week 1)
PB-02	Database Design & Setup	Database for users, products, transactions, inventory, finance	High	March (Week 1)
PB-03	User Authentication & Roles	Login using email/username with role-based access (Admin, Cashier, Owner)	High	March (Week 1)
PB-04	POS Transaction Module	Process sales transactions, calculate totals, taxes, and discounts	High	March (Week 2-3)

PRODUCT BACKLOG

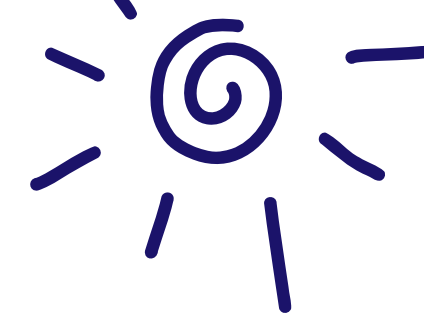


ID	Backlog Item	Description	Priority	Target Period
PB-05	Receipt Generation	Generate and print/digital receipts after transactions	High	March (Week 3)
PB-06	Product Management	Add, edit, delete products with price and SKU	High	March (Week 4)
PB-07	Inventory Management	Track stock in/out automatically based on transactions	High	April (Week 1)
PB-08	Low Stock Alert	Notify admin when inventory reaches minimum threshold	Medium	April (Week 2-3)





PRODUCT BACKLOG



ID	Backlog Item	Description	Priority	Target Period
PB-09	Supplier Management	Manage supplier data for inventory restocking	Medium	April (Week 2)
PB-10	Purchase & Restock Module	Record inventory restock and update stock levels	High	April (Week 2-3)
PB-11	Financial Report – Sales	Generate daily, weekly, and monthly sales reports	High	April (Week 3)
PB-12	Financial Report – Profit & Loss	Automatically calculate revenue, cost, and profit	High	April (Week 4)





PRODUCT BACKLOG

ID	Backlog Item	Description	Priority	Target Period
PB-13	Cash Flow Report	Track cash inflow and outflow	Medium	May (Week 1)
PB-14	Dashboard Analytics	Visual dashboard showing sales, stock, and profit summary	High	May (Week 1-2)
PB-15	Export Reports	Export reports to PDF/Excel	Medium	May (Week 2)
PB-16	Security & Data Validation	Secure transactions, input validation, access control	Medium	May (Week 3)
PB-17	Final Integration Testing & Bug Fixing	Integration testing and bug fixing	High	May (Week 3-4)
PB-18	Deployment & Documentation	Deploy system and prepare user documentation	Medium	May (Week 4)





BUSINESS CASE

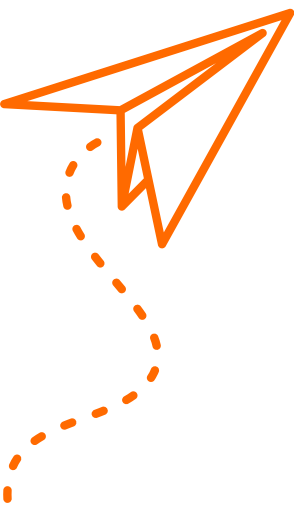


SnapCash: Quick, Easy, Snap



All-in-One POS & Business Management

- Transactions, inventory, reporting, analytics
- Role-based access (owner, cashier, admin)
- Subscription tiers:
 - Standard Rp 100K/mo
 - Premium Rp 300K/mo
 - Enterprise Rp 1.5M/mo



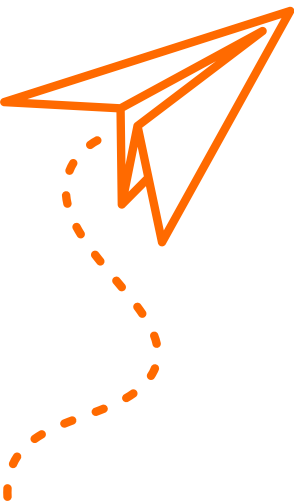


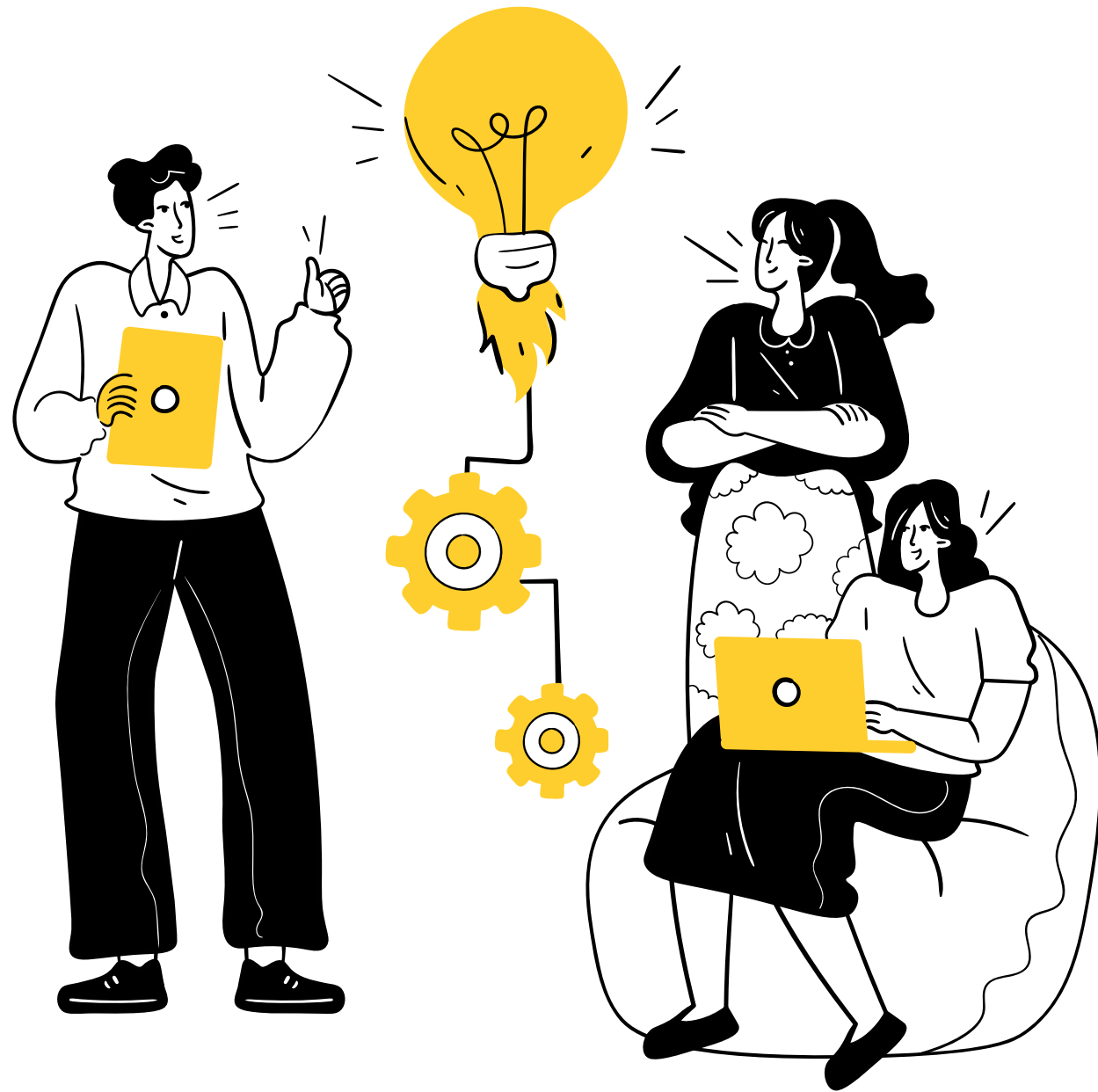
Financial Snapshot

- Year 1: 100 users → Rp 180M revenue (loss Rp 90M)
- Year 3: 1,000 users + AI → Rp 950M profit
- Year 5: 3,000 users → Rp 5.4B revenue, Rp 3.2B profit
-

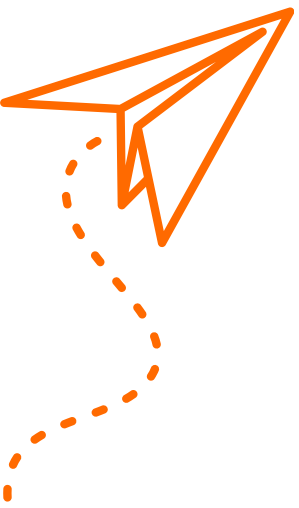
Vision

From POS → Business Intelligence Partner for MSMEs
Bank/fintech integration • Open API • ASEAN
expansion





PROJECT CHARTER



A. PROJECT GENERAL INFORMATION



Project Name: SnapCash – POS and Business Management System

Project Description

SnapCash is a digital Point of Sale (POS) system designed to help small and medium-sized businesses, especially in the food and retail sectors, manage their daily operations efficiently. The system provides features such as transaction processing, inventory management, financial reporting, and business analytics in one integrated platform. By using SnapCash, businesses can digitize their operations, reduce manual errors, and gain insights into their financial performance.

Project Duration

1 March 2026 – 31 May 2026

Total Sprint

6 Sprints (Each Sprint = 2 Weeks)

B. ROLE ASSIGNING

Product Owner: Angelo

Scrum Master: Kennes Jansen

Development Team

- Christofer Marvelous Krisdianto
- Nabila Ullivia Fatimah
- Meini Rusiadi



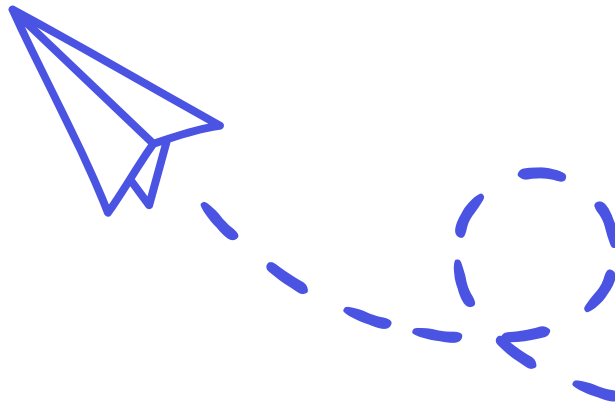
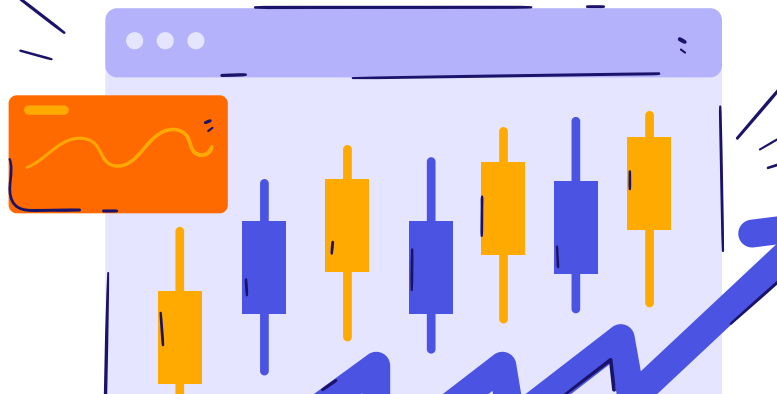
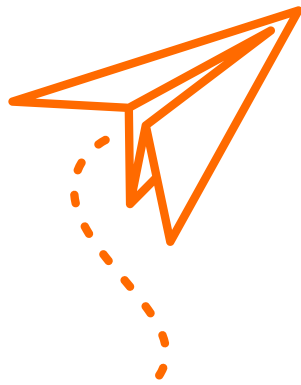
C. SPRINT PLAN

Sprint 1
Duration:
1 March – 14 March 2026

Sprint Goal:
Establish system foundation including UI design, database setup, and authentication.

Sprint Backlog

ID	Task	Story Point	Assigned To
PB-01	UI/UX Design Prototype	5	Nabila Ullivia Fatimah, Meini Rusiadi
PB-02	Database Design & Setup	5	Christofer Krisdianto Marveleous
PB-03	User Authentication & Roles	5	Christofer Krisdianto Marveleous



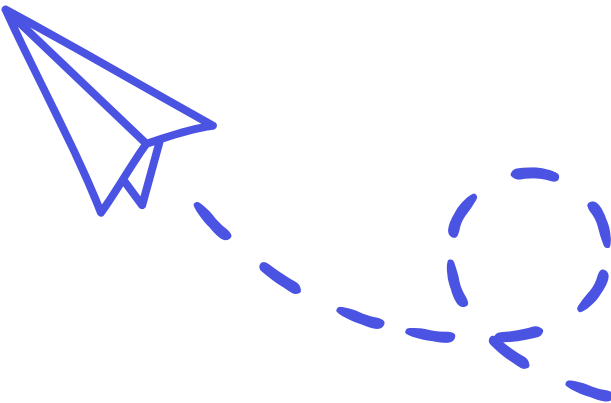
C. SPRINT PLAN

Sprint 2
Duration:
15 March – 28 March 2026
Sprint Goal:
Develop the core POS transaction system.
Sprint Backlog

ID	Task	Story Point	Assigned To
PB-04	POS Transaction Module	8	Christofer Krisdianto Marveleous
PB-05	Receipt Generation	3	Meini Rusiadi
PB-06	Product Management	5	Nabila Ullivia Fatimah

C. SPRINT PLAN

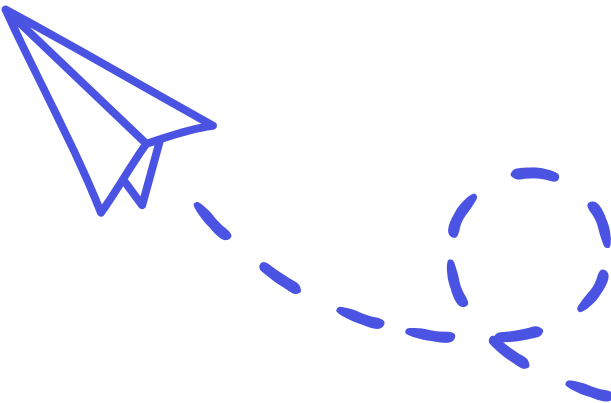
Sprint 3
Duration:
29 March – 11 April 2026
Sprint Goal:
Develop inventory tracking and supplier management.
Sprint Backlog



ID	Task	Story Point	Assigned To
PB-07	Inventory Management	5	Christofer Krisdianto Marveleous
PB-08	Low Stock Alert	3	Nabila Ullivia Fatimah
PB-09	Supplier Management	8	Christofer Krisdianto, Marveleous Nabila Ullivia Fatimah
PB-10	Purchase & Restock Module	5	Meini Rusiadi

C. SPRINT PLAN

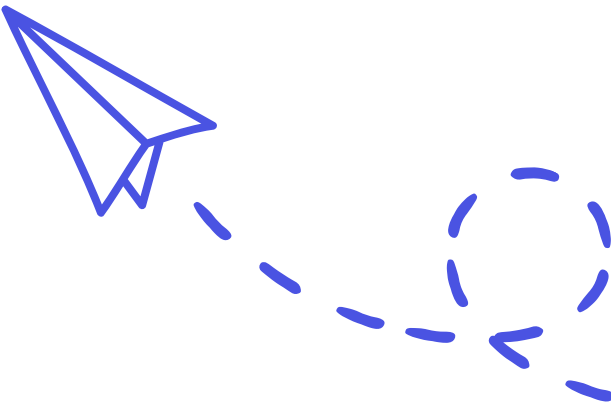
Sprint 4
Duration:
12 April – 25 April 2026
Sprint Goal:
Develop financial reporting features.
Sprint Backlog



ID	Task	Story Point	Assigned To
PB-11	Financial Report – Sales	8	Christofer Krisdianto Marveleous
PB-12	Financial Report – Profit & Loss	5	Meini Rusiadi
PB-13	Cash Flow Report	5	Nabila Ullivia Fatimah

C. SPRINT PLAN

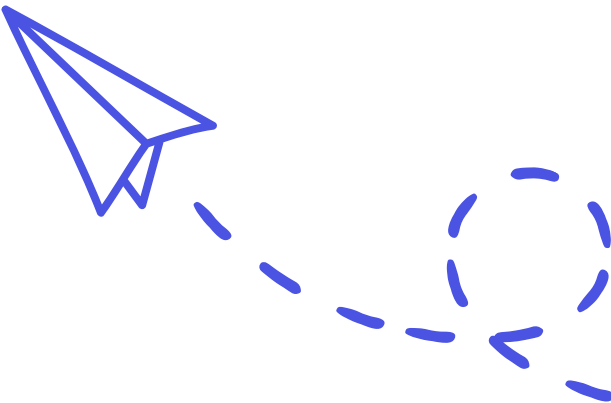
Sprint 5
Duration:
26 April – 9 May 2026
Sprint Goal:
Develop business analytics and report exporting features.
Sprint Backlog



ID	Task	Story Point	Assigned To
PB-14	Dashboard Analytics	5	Meini Rusiadi
PB-15	Export Reports	5	Nabila Ullivia Fatimah
PB-16	Security & Data Validation	5	Christofer Krisdianto Marveleous

C. SPRINT PLAN

Sprint 6
Duration:
10 May – 23 May 2026
Sprint Goal:
Finalize system development and prepare deployment.
Sprint Backlog



ID	Task	Story Point	Assigned To
PB-17	System Testing & Bug Fixing	8	Christofer Krisdianto, Marveleous Meini Rusiadi
PB-18	Deployment & Documentation	5	Nabila Ullivia Fatimah

D. FINAL PHASE

24 May – 31 May 2026

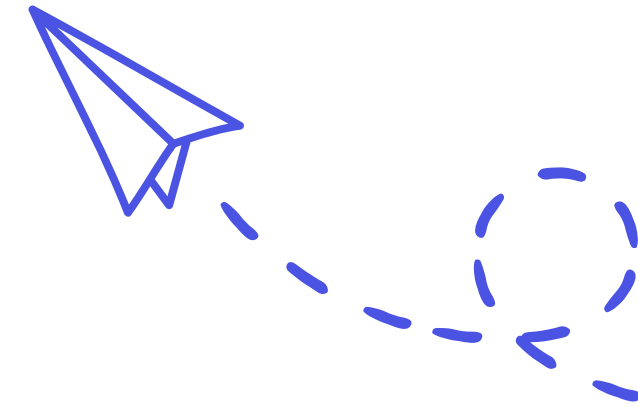
Activities:

- Final system testing
- Deployment preparation
- User documentation creation
- Project evaluation
- Final presentation preparation

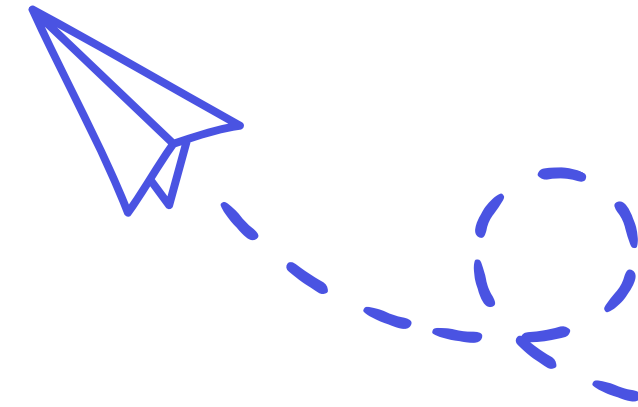
Definition of Done (DoD)

A backlog item is considered complete when:

- The feature has been implemented successfully
- The feature passes testing without major bugs
- The system integrates with other modules correctly
- Code has been committed to the repository
- The feature is ready to be demonstrated in Sprint Review



SPRINT EXECUTION PLAN



Sprint 1

Duration: 1 March – 14 March 2026

Sprint Focus

Establish the foundation of the system, including UI/UX design, database setup, and user authentication.

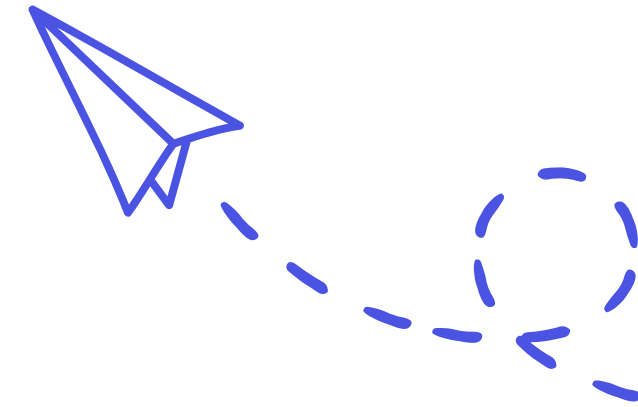
Activities

- Design POS cashier-friendly interface prototype
- Create database schema for users, products, transactions, inventory, and finance
- Setup development environment and repository
- Implement login system with role-based access (Admin, Cashier, Owner)
- Integrate authentication with database

Deliverables

- UI/UX POS Prototype
- Database structure and tables
- Functional login system with role-based access
- Initial system architecture ready for development

SPRINT EXECUTION PLAN



Sprint 2

Duration: 15 March – 28 March 2026

Sprint Focus

Develop the core POS transaction system and product management features.

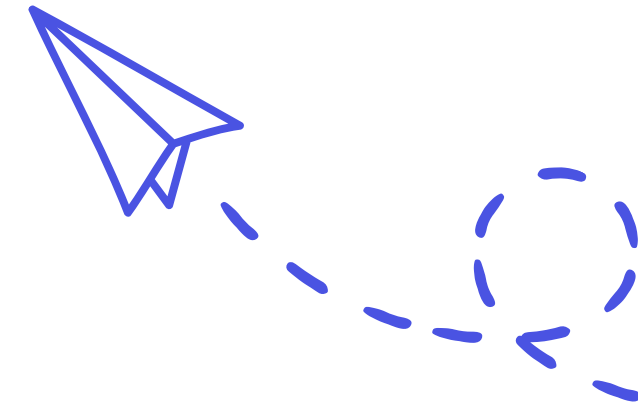
Activities

- Implement POS transaction processing module
- Develop product management feature (add/edit/delete products)
- Integrate product database with POS system
- Implement receipt generation for completed transactions
- Conduct initial testing for transaction flow

Deliverables

- Functional POS transaction module
- Product management system
- Digital receipt generation
- Tested transaction workflow

SPRINT EXECUTION PLAN



Sprint 3

Duration: 29 March – 11 April 2026

Sprint Focus

Develop inventory management and supplier management features.

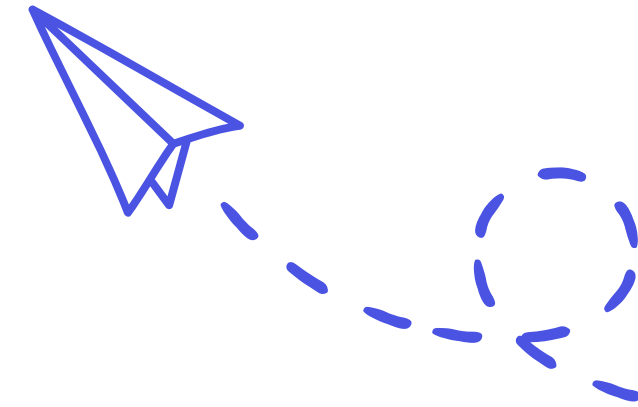
Activities

- Implement inventory tracking system
- Integrate automatic stock deduction during transactions
- Develop low stock alert notification system
- Create supplier management module
- Develop purchase and restock module for inventory updates

Deliverables

- Inventory management system
- Low stock alert feature
- Supplier management module
- Purchase and restock functionality

SPRINT EXECUTION PLAN



Sprint 4

Duration: 12 April – 25 April 2026

Sprint Focus

Develop financial reporting features to track business performance.

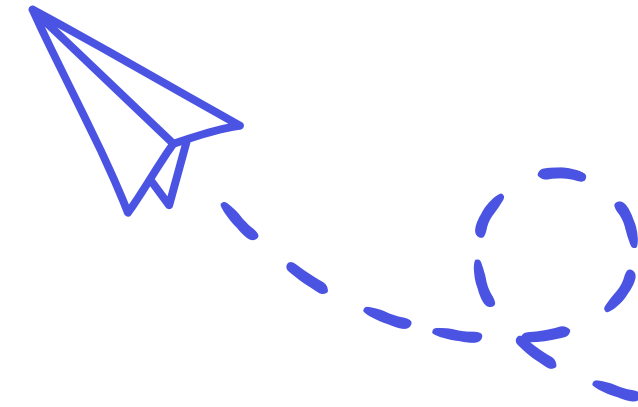
Activities

- Implement daily, weekly, and monthly sales reports
- Develop profit and loss calculation system
- Implement cash flow tracking functionality
- Integrate financial data with transaction and inventory systems
- Test report accuracy and data consistency

Deliverables

- Sales report system
- Profit and Loss reporting feature
- Cash flow report feature
- Integrated financial reporting module

SPRINT EXECUTION PLAN



Sprint 5

Duration: 26 April – 9 May 2026

Sprint Focus

Develop business analytics dashboard and reporting features.

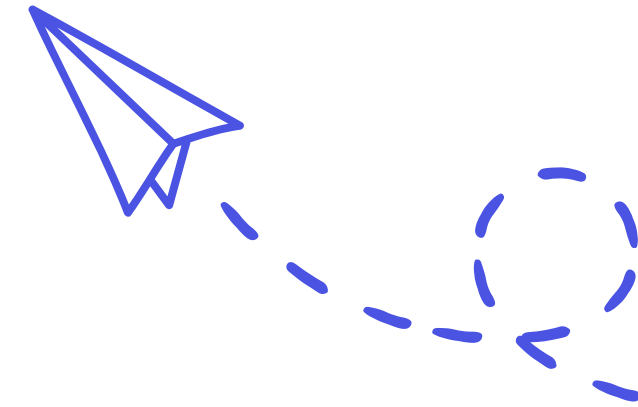
Activities

- Design and develop dashboard analytics interface
- Implement sales, inventory, and financial summary visualization
- Develop export functionality for reports (PDF/Excel)
- Implement security features and input validation
- Improve system usability and user interface

Deliverables

- Dashboard analytics interface
- Exportable reports (PDF/Excel)
- Security and data validation system
- Improved user interface and usability

SPRINT EXECUTION PLAN



Sprint 6

Duration: 10 May – 23 May 2026

Sprint Focus

Finalize system development through testing, debugging, deployment preparation, and documentation.

Activities

- Conduct full system integration testing
- Identify and fix system bugs and errors
- Optimize system performance
- Prepare deployment environment
- Create user documentation and system guide

Deliverables

- Fully integrated SnapCash system
- Bug fixes and optimized system performance
- Deployed system ready for use
- User documentation and deployment report

FINAL PROJECT ACTIVITIES

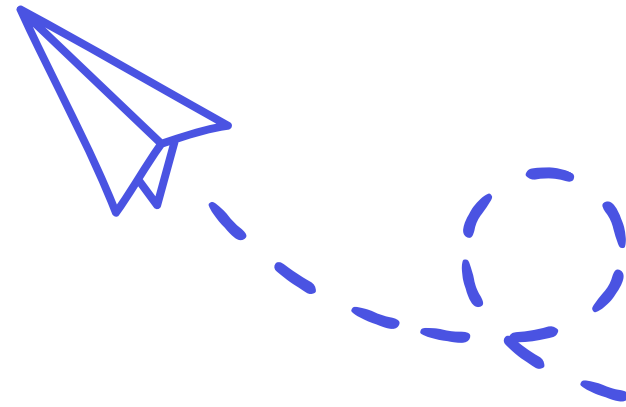
24 May – 31 May 2026

Activities:

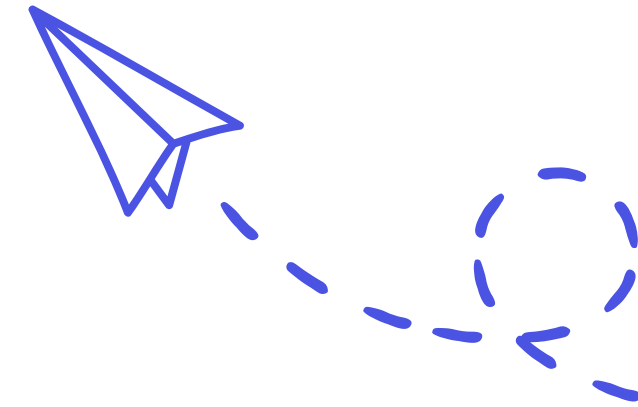
- Final system review
- Product demonstration
- Documentation completion
- Final presentation preparation

Deliverables:

- Final SnapCash system
- Complete project documentation
- Final project presentation



DAILY SCRUM LOG - SNAPCASH PROJECT



Daily Scrum Duration: 45 Minutes

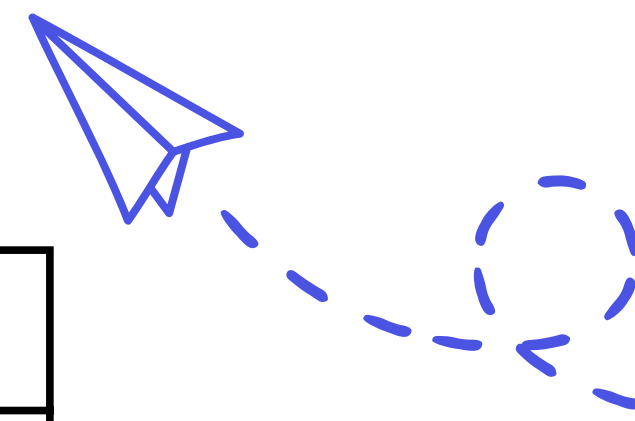
Meeting Time: 09.00 AM – 09.45 AM

Participants:

- Product Owner
- Scrum Master
- Development Team

SPRINT 1 DAILY SCRUM LOG

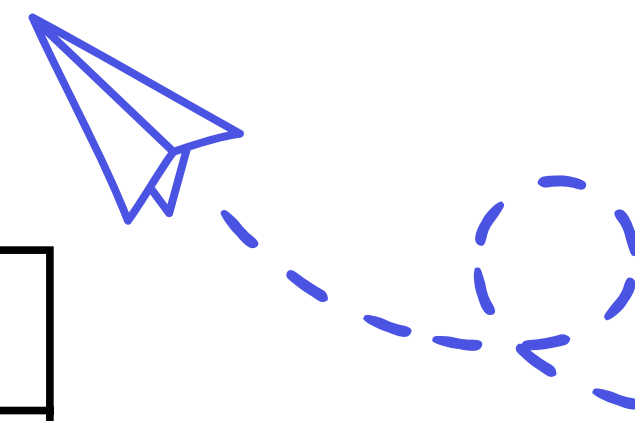
Sprint Duration: 1 – 14 March 2026
Sprint Focus: System foundation (UI/UX, Database, Authentication)



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning and requirement discussion	Setup project repository and development tools	None
Day 2	Created initial UI wireframe	Continue POS UI prototype design	None
Day 3	Completed POS interface design draft	Begin database schema design	None
Day 4	Designed database schema	Implement database tables	Minor schema adjustments
Day 5	Database setup completed	Start login authentication system	None
Day 6	Developed login interface	Implement role-based access control	None
Day 7	Integrated login system with database	Testing authentication system	None
Day 8	Fixed login validation issues	Prepare system for next module	None
Day 9	Final authentication testing	Documentation for sprint deliverables	None
Day 10	Sprint review preparation	Sprint retrospective	None

SPRINT 2 DAILY SCRUM LOG

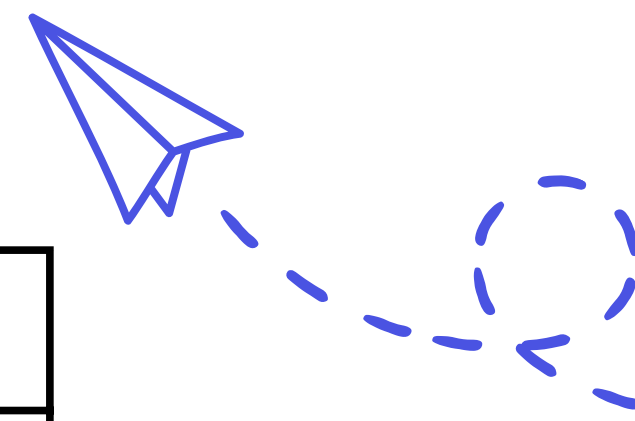
Sprint Duration: 15 – 28 March 2026
Sprint Focus: POS transaction and product management



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design POS transaction flow	None
Day 2	POS workflow designed	Implement transaction logic	None
Day 3	Transaction module development	Continue transaction processing	None
Day 4	Product database integration	Build product management interface	None
Day 5	Product CRUD feature implemented	Integrate POS with product list	None
Day 6	Transaction testing	Implement receipt generation	None
Day 7	Receipt module development	Continue receipt formatting	None
Day 8	Transaction & receipt integration	Fix calculation issues	Minor bugs
Day 9	System testing	Prepare sprint deliverables	None
Day 10	Sprint review preparation	Sprint retrospective	None

SPRINT 3 DAILY SCRUM LOG

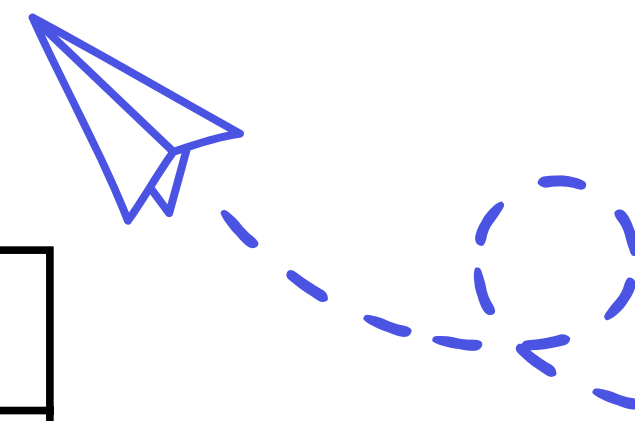
Sprint Duration: 29 March – 11 April 2026
Sprint Focus: Inventory and supplier management



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design inventory structure	None
Day 2	Inventory database tables	Implement stock tracking logic	None
Day 3	Automatic stock deduction feature	Testing inventory updates	None
Day 4	Inventory testing	Implement low stock alert	None
Day 5	Low stock alert notification	Supplier management design	None
Day 6	Supplier database setup	Supplier CRUD features	None
Day 7	Purchase/restock module	Integrate supplier system	None
Day 8	Inventory restock testing	Fix stock update bugs	Minor bug
Day 9	System integration testing	Documentation	None
Day 10	Sprint review preparation	Sprint retrospective	None

SPRINT 4 DAILY SCRUM LOG

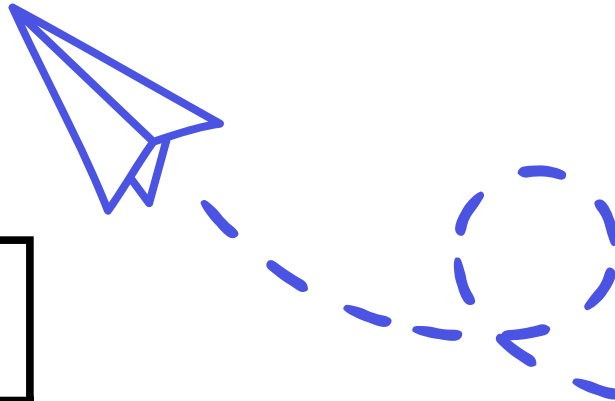
Sprint Duration: 12 – 25 April 2026
Sprint Focus: Financial reporting system



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design reporting structure	None
Day 2	Sales report development	Integrate transaction data	None
Day 3	Sales report completed	Develop profit & loss module	None
Day 4	Profit & loss calculations	Testing financial accuracy	None
Day 5	Fix calculation errors	Develop cash flow report	None
Day 6	Cash flow module development	Integrate reports with dashboard	None
Day 7	Report interface improvements	Testing report generation	None
Day 8	Financial report debugging	Data validation	None
Day 9	System testing	Prepare sprint review	None
Day 10	Sprint review preparation	Sprint retrospective	None

SPRINT 5 DAILY SCRUM LOG

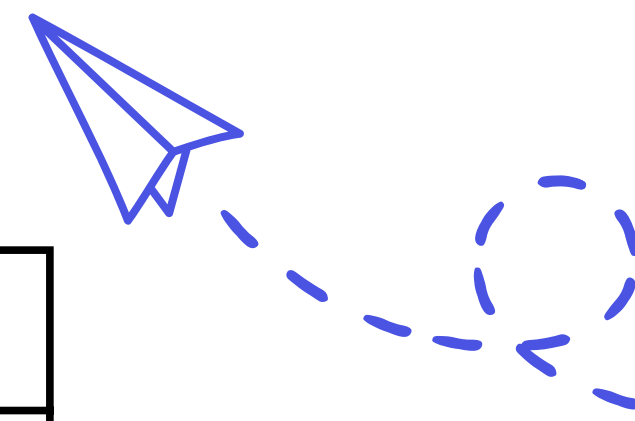
Sprint Duration: 26 April – 9 May 2026
Sprint Focus: Dashboard analytics and reporting export



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Design dashboard UI	None
Day 2	Dashboard prototype	Implement data visualization	None
Day 3	Sales analytics chart	Integrate inventory analytics	None
Day 4	Dashboard integration	Develop export report feature	None
Day 5	PDF export system	Excel export system	None
Day 6	Export testing	Implement security validation	None
Day 7	Access control improvements	Input validation testing	None
Day 8	Dashboard usability improvements	Fix UI bugs	Minor UI issues
Day 9	Final dashboard testing	Documentation	None
Day 10	Sprint review preparation	Sprint retrospective	None

SPRINT 6 DAILY SCRUM LOG

Sprint Duration: 10 – 23 May 2026
Sprint Focus: System testing, deployment, documentation



Day	Yesterday Progress	Today Plan	Blockers
Day 1	Sprint planning	Begin system integration testing	None
Day 2	Integration testing	Identify system bugs	None
Day 3	Fix transaction module bugs	Continue debugging	None
Day 4	Fix inventory bugs	Optimize system performance	None
Day 5	Performance testing	Security testing	None
Day 6	Security improvement	Prepare deployment environment	None
Day 7	Deployment preparation	Create user documentation	None
Day 8	Documentation writing	Final system testing	Minor UI issues
Day 9	Final bug fixes	Prepare final presentation	None
Day 10	Sprint review preparation	Sprint retrospective	None

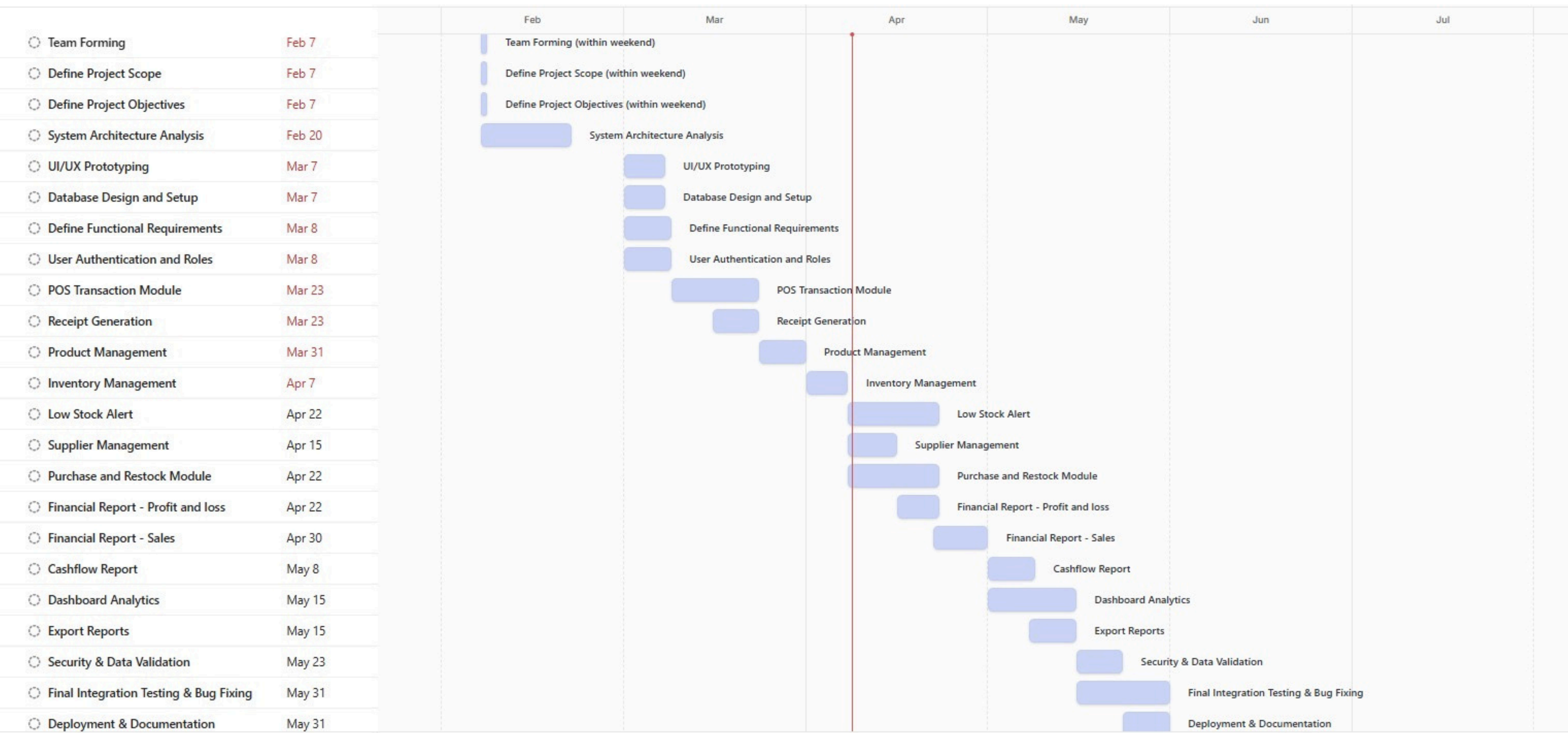
Work Breakdown Structure

Project Title	SnapCash				
Team Leader	Angelo				
Company Name	SnapCash				
Date	28/02/2026				
WBS Number	Task Title	Persons	Start Date	Due Date	Progress
1	PROJECT INITIATION & PLANNING				
1.1	Team Forming	All Member ▼	07/02/2026	07/02/2026	Done ▼
1.2	Define Project Scope	All Member ▼	07/02/2026	07/02/2026	Done ▼
1.3	Define Project Objectives	Angelo ▼	07/02/2026	07/02/2026	Done ▼
Deliverables	Project Scope and Objectives				
2	REQUIREMENTS GATHERING & ANALYSIS				
2.1	System Architecture Analysis	Abil ▼	07/02/2026	20/02/2026	Done ▼
2.2	UI/UX Prototyping	Meini ▼	01/03/2026	07/03/2026	Done ▼
2.3	Database Design and Setup	Kennes ▼	01/03/2026	08/03/2026	Done ▼
2.4	Define Functional Requirements	Kennes ▼	01/03/2026	08/03/2026	Done ▼
Deliverables	System Architecture, UI/UX Prototype, Database Setup, Functional Requirements				
3	SYSTEM & DATA MODELLING				
3.1	User Authentication and Roles	Marvel ▼	01/03/2026	08/03/2026	To Do ▼
3.2	POS Transaction Module	Marvel ▼	09/03/2026	23/03/2026	To Do ▼
3.3	Receipt Generation	Marvel ▼	16/03/2026	23/03/2026	To Do ▼
3.4	Product Management	Marvel ▼	24/03/2026	31/03/2026	To Do ▼
3.5	Inventory Management	Kennes ▼	01/04/2026	07/04/2026	To Do ▼
3.6	Low Stock Alert	Kennes ▼	08/04/2026	22/04/2026	To Do ▼
3.7	Supplier Management	Kennes ▼	08/04/2026	15/04/2026	To Do ▼
3.8	Purchase & Restock Module	Marvel ▼	08/04/2026	22/04/2026	To Do ▼
3.9	Financial Report - Profit and Loss	Marvel ▼	16/04/2026	22/04/2026	To Do ▼
3.10	Financial Report- Sales	Abil ▼	22/04/2026	30/04/2026	To Do ▼
3.11	Cashflow Report	Abil ▼	01/05/2026	08/05/2026	To Do ▼
3.12	Dashboard Analytics	Meini ▼	01/05/2026	15/05/2026	To Do ▼
3.13	Export Reports	Meini ▼	08/05/2026	15/05/2026	To Do ▼
Deliverables	Usable and Ready to Test SnapCash App				To Do ▼
6	Testing				To Do ▼
4.1	Security & Data validation	Angelo ▼	16/05/2026	23/05/2026	To Do ▼
4.2	Final Integration Testing & Bug Fixing	Angelo ▼	16/05/2026	31/05/2026	To Do ▼
4.3	Deployment & Documentation	Angelo ▼	24/05/2026	31/05/2026	To Do ▼
Deliverables	Ready to use SnapCash App				

SNAPCASH PROJECT BUDGET PLAN

Category	Amount (IDR)	%	Breakdown
Human Resources	Rp 75,000,000	37.5%	5 developers × 5 months @ Rp3M/person/month
Infrastructure & Hosting	Rp 30,000,000	15%	Cloud server, database, domain & SSL — Rp6M/month × 5
Software & Tools	Rp 25,000,000	12.5%	Development licenses, APIs, third-party subscriptions
Hardware / Equipment	Rp 25,000,000	12.5%	POS terminal, barcode scanner, receipt printer for testing
UI/UX Design	Rp 15,000,000	7.5%	Figma licenses, design sprints, prototyping & assets
Testing & QA	Rp 10,000,000	5%	Manual & automated testing, bug fixes, UAT sessions
Documentation & Training	Rp 10,000,000	5%	User manuals, onboarding materials, handover docs
Contingency Reserve	Rp 10,000,000	5%	Buffer for unexpected costs — only use if overrun occurs
TOTAL	Rp 200,000,000	100%	Full project budget — 4 to 6 months

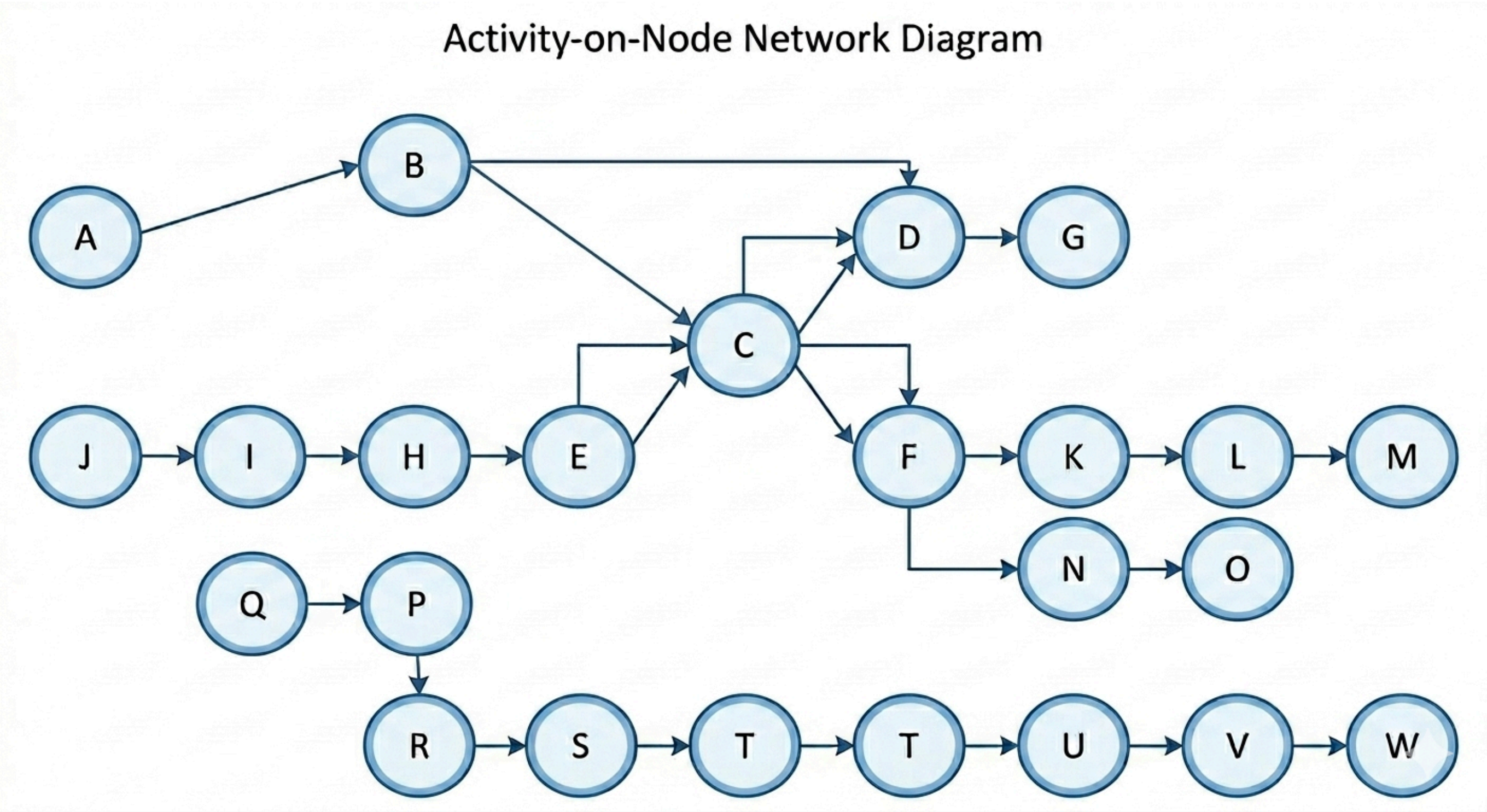
GANTT CHART



ACTIVITY ON NODE

Task	Task Title	Duration	Predecessor
A	Team Forming	1	None
B	Define Project Scope	1	A
C	Define Project Objectives	1	A
D	System Architecture Analysis	14	B,C
E	UI/UX Prototyping	7	B,C
F	Database Design and Setup	7	B,C
G	Define Functional Requirements	7	D
H	User Authentication and Roles	7	E,F
I	POS Transaction Module	14	E,F
J	Receipt Generation	7	I
K	Product Management	7	E,F
L	Inventory Management	7	K
M	Low Stock Alert	14	L
N	Supplier Management	7	E,F
O	Purchase & Restock Module	14	N
P	Financial Report - Profit and Loss	7	I
Q	Financial Report- Sales	7	I
R	Cashflow Report	7	P,Q
S	Dashboard Analytics	14	P,Q,R
T	Export Reports	7	S
U	Security & Data validation	7	T
V	Final Integration Testing & Bug Fixing	14	U
W	Deployment & Documentation	7	V

ACTIVITY ON NODE DIAGRAM



JALUR 1:

$$\text{A-B/C-D-G} = 1+1+14+7 = 23 \text{ HARI}$$

JALUR 2:

$$\text{A-B/C-E/F-H} = 1+1+7+7 = 16 \text{ HARI}$$

JALUR 3:

$$\text{A-B/C-E/F-I-J} = 1+1+7+14+7 = 30 \text{ HARI}$$

JALUR 4:

$$\text{A-B/C-E/F-K-L-M} = 1+1+7+7+7+14 = 37 \text{ HARI}$$

JALUR 5:

$$\text{A-B/C-E/F-N-O} = 1+1+7+7+14 = 30 \text{ HARI}$$

JALUR 6:

$$\text{A-B/C-E/F-I-P/Q-R-S-T-U-V-W}$$

$$= 1+1+7+14+7+7+14+7+7+14+7 = 86 \text{ HARI}$$

FORWARD PASS

Task	ES	EF	Ket
A	0	1	Mulai dari 0
B	1	2	EF A jadi ES B
C	1	2	EF A jadi ES C
D	2	16	Max EF B,C
E	2	9	Max EF B,C
F	2	9	Max EF B,C
G	16	23	EF D
H	9	16	Max EF E,F
I	9	23	Max EF E,F
J	23	30	EF I
K	9	16	Max EF E,F
L	16	23	EF K
M	23	37	EF L
N	9	16	Max EF E,F
O	16	30	EF N
P	23	30	EF I
Q	23	30	EF I
R	30	37	Max EF P,Q
S	37	51	Max EF R
T	51	58	EF S
U	58	65	EF T
V	65	79	EF U
W	79	86	Proyek selesai 86 HARI

BACKWARD PASS

Task	Successor	LF	LS	Ket
W	None	86	79	Start from project duration
V	W	79	65	
U	V	65	58	
T	U	58	51	
S	T	51	37	
R	S	37	30	
P	R	30	23	
Q	R	30	23	
O	None	86	72	
N	O	72	65	
M	None	86	72	
L	M	72	65	
K	L	65	58	
J	None	86	79	
I	J,P,Q	23	9	
H	None	86	79	
G	None	86	79	
F	H,I,K,N	9	2	
E	H,I,K,N	9	2	
D	G	79	65	
C	D,E,F	2	1	
B	D,E,F	2	1	
A	B,C	1	0	

SLACK / FLOAT (LS – ES)

$$A = 0 - 0 = 0 \text{ (KRITIS)}$$

$$B = 1 - 1 = 0 \text{ (KRITIS)}$$

$$C = 1 - 1 = 0 \text{ (KRITIS)}$$

$$D = 65 - 2 = 63$$

$$E = 2 - 2 = 0 \text{ (KRITIS)}$$

$$F = 2 - 2 = 0 \text{ (KRITIS)}$$

$$G = 79 - 16 = 63$$

$$H = 79 - 9 = 70$$

$$I = 9 - 9 = 0 \text{ (KRITIS)}$$

$$J = 79 - 23 = 56$$

$$K = 58 - 9 = 49$$

$$L = 65 - 16 = 49$$

$$M = 72 - 23 = 49$$

$$N = 65 - 9 = 56$$

$$O = 72 - 16 = 56$$

$$P = 23 - 23 = 0 \text{ (KRITIS)}$$

$$Q = 23 - 23 = 0 \text{ (KRITIS)}$$

$$R = 30 - 30 = 0 \text{ (KRITIS)}$$

$$S = 37 - 37 = 0 \text{ (KRITIS)}$$

$$T = 51 - 51 = 0 \text{ (KRITIS)}$$

$$U = 58 - 58 = 0 \text{ (KRITIS)}$$

$$V = 65 - 65 = 0 \text{ (KRITIS)}$$

$$W = 79 - 79 = 0 \text{ (KRITIS)}$$

RISK MANAGEMENT PLAN



RISK ASSESSMENT METHODOLOGY

Risk Level	Score	Description
CRITICAL	High × High	Immediate action required; escalate to Scrum Master
HIGH	Med-High	Develop detailed mitigation plan within current sprint
MEDIUM	Medium	Monitor weekly; address if trend worsens
LOW	Low	Accept and monitor passively

RISK PROBABILITY & IMPACT MATRIX

Impact ↓ / Likelihood →	Low	Medium	High
Low	LOW R-05, R-14	LOW	MEDIUM
Medium	LOW	MEDIUM R-04, R-08 R-11, R-13	HIGH R-02, R-03 R-12, R-15
High	MEDIUM	HIGH R-10	CRITICAL R-01, R-06 R-07, R-09

RISK REGISTER

ID	Category	Risk Description	Likelihood	Impact	Level	Mitigation Strategy
R-01	Technical	Integration bugs between POS, inventory, and financial modules	High	High	CRITICAL	Conduct integration testing every sprint; use a shared API contract between modules.
R-02	Technical	Database schema errors causing data loss or inconsistency	Medium	High	HIGH	Implement database migration scripts; conduct schema review before Sprint 2.
R-03	Technical	Authentication/role-based access vulnerability	Medium	High	HIGH	Use JWT token + bcrypt hashing; conduct security audit in Sprint 5 (PB-16).
R-04	Technical	Low performance when processing many concurrent transactions	Medium	Medium	MEDIUM	Optimize database queries; add indexing on critical fields; load-test before deployment.
R-05	Technical	Export report (PDF/Excel) format incompatibility	Low	Medium	LOW	Use well-tested libraries (e.g., jsPDF, SheetJS); test on multiple devices.
R-06	Schedule	Sprint delay due to underestimated story points	High	High	CRITICAL	Add buffer time each sprint; hold daily standups; re-estimate in Sprint Review if needed.
R-07	Schedule	System Testing & Bug Fixing (PB-17) underscoped, pushing beyond deadline	High	High	CRITICAL	Begin regression testing from Sprint 4 onward; not only in Sprint 6.
R-08	Schedule	Late deployment causes incomplete user documentation	Medium	Medium	MEDIUM	Prepare documentation template early; assign dedicated member (Nabila) from Sprint 5.
R-09	Schedule	Key developer (Christofer) overloaded – assigned to 6+ backlog items	High	High	CRITICAL	Redistribute tasks; pair-program critical features; Scrum Master monitors workload.

RISK REGISTER

ID	Category	Risk Description	Likelihood	Impact	Level	Mitigation Strategy
R-10	Team	Team member absence or dropout mid-project	Low	High	HIGH	Document all code with comments; use Git for version control; cross-train team members.
R-11	Team	Miscommunication between Product Owner and Development Team	Medium	High	MEDIUM	Hold Sprint Planning & Review meetings consistently; use shared backlog in project management tool.
R-12	Scope	Scope creep – adding features beyond original backlog	Medium	High	HIGH	Strictly follow Definition of Done; Product Owner (Angelo) approves all scope changes.
R-13	Scope	Dashboard Analytics (PB-14) requirements unclear or too complex	Medium	Medium	MEDIUM	Define specific KPIs/metrics for dashboard in Sprint 4; create wireframe before coding.
R-14	External	Third-party library deprecation or breaking change (e.g., POS UI library)	Low	Medium	LOW	Lock dependency versions; monitor changelog; use widely-supported libraries.
R-15	External	MSME user adoption lower than expected due to UI complexity	Medium	High	HIGH	Conduct usability testing with real MSME users in Sprint 5; iterate based on feedback.

RISK OWNERSHIP & MONITORING

ID	Risk Description	Owner	Review	Contingency Plan
R-01	Integration bugs between modules	Christofer (Lead Dev)	Every Sprint	Roll back to last stable version; isolate failing module.
R-02	Database schema errors	Christofer	Sprint 1–2	Restore from backup; apply migration rollback script.
R-03	Authentication vulnerability	Christofer	Sprint 5	Disable affected endpoint; hotfix within 24 hours.
R-04	Low performance on concurrent transactions	Christofer	Sprint 5–6	Reduce query complexity; cache frequently accessed data.
R-05	Export report incompatibility	Nabila	Sprint 5	Switch to alternative library; manual export fallback.
R-06	Sprint delay	Kennes (Scrum Master)	Every Sprint	Reprioritize backlog; defer low-priority items to next sprint.
R-07	Testing underscoped	Kennes (Scrum Master)	Sprint 4–6	Extend Sprint 6; reduce scope of non-critical features.
R-08	Incomplete documentation	Nabila	Sprint 5–6	Use auto-documentation tools; focus on critical user flows.
R-09	Christofer overloaded	Kennes (Scrum Master)	Every Sprint	Pair-program with Meini; redistribute tasks in Sprint Planning.

RISK OWNERSHIP & MONITORING

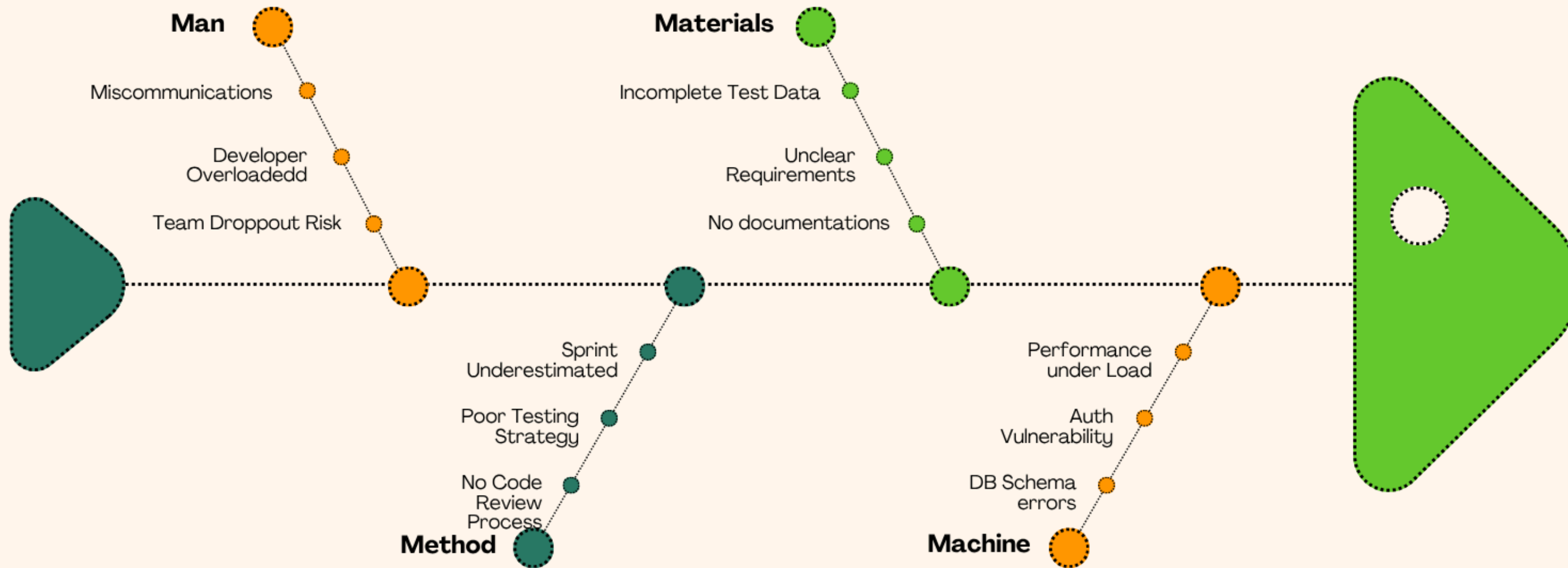
ID	Risk Description	Owner	Review	Contingency Plan
R-10	Team member dropout	Angelo (PO)	Ongoing	Redistribute tasks; prioritize critical features only.
R-11	Miscommunication PO ↔ Dev Team	Angelo (PO)	Every Sprint	Re-conduct Sprint Planning; clarify acceptance criteria.
R-12	Scope creep	Angelo (PO)	Every Sprint	Freeze scope; log new requests for post-project phase.
R-13	Dashboard Analytics unclear requirements	Meini	Sprint 4	Simplify to 3 core KPIs: daily revenue, stock level, profit.
R-14	Third-party library breaking change	Christofer	Sprint 2, 5	Pin dependency versions in package.json; test after updates.
R-15	Low MSME user adoption	Angelo (PO)	Sprint 5–6	Redesign critical workflows; provide onboarding walkthrough.

RISK SUMMARY BY SPRINT

Sprint	Period	Key Risks to Watch	Recommended Action
Sprint 1	1–14 Mar	R-02 (DB Schema), R-03 (Auth), R-09 (Overload)	Define DB schema carefully; peer-review before finalizing.
Sprint 2	15–28 Mar	R-01 (Integration), R-06 (Delay), R-09 (Overload)	Set API contract between POS & Inventory modules early.
Sprint 3	29 Mar–11 Apr	R-01, R-06, R-12 (Scope Creep)	Strictly enforce DoD; avoid adding new inventory features.
Sprint 4	12–25 Apr	R-11 (Miscommunication), R-13 (Dashboard Scope)	Clarify financial KPIs with PO before development starts.
Sprint 5	26 Apr–9 May	R-03 (Security), R-15 (Usability), R-07 (Testing)	Start usability tests; run security <u>audit</u> on authentication.
Sprint 6	10–23 May	R-07 (Testing Scope), R-08 (Documentation), R-06	Prioritize bug fixes; complete documentation in parallel.

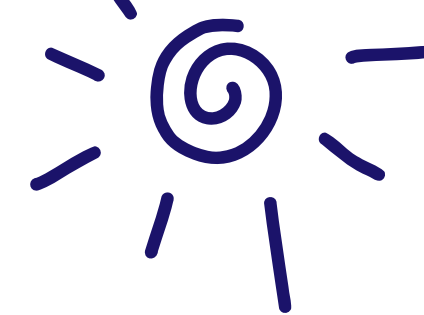
SnapCash

Fishbone Diagram



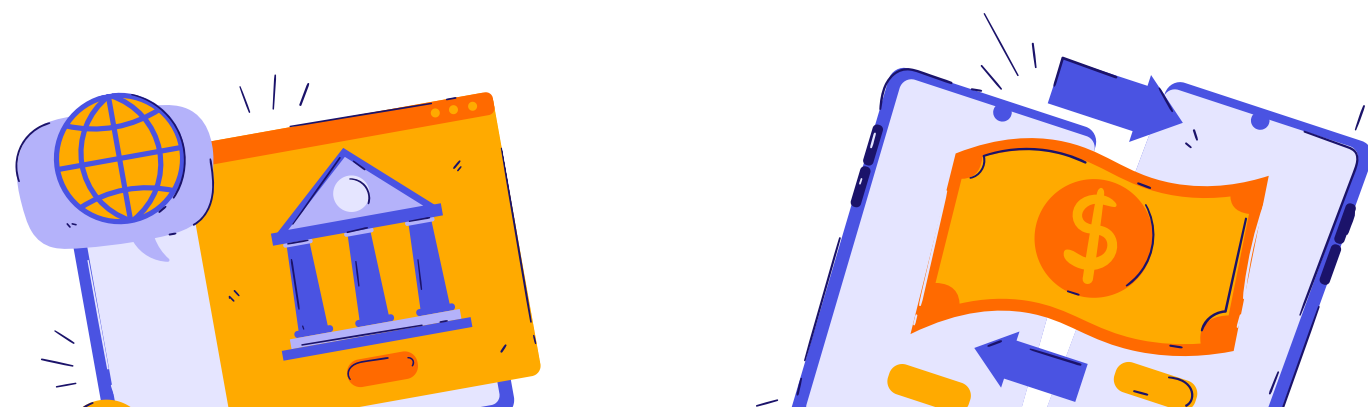
COMMUNICATION PLAN





PURPOSE OF THE COMMUNICATION PLAN

The Communication Management Plan for the SnapCash project defines how information is distributed to each stakeholder throughout the project lifecycle. It ensures that every stakeholder receives the right information, in the right format, at the right time, through the appropriate contact person. The plan covers all four phases of the project: Project Initiation and Planning, Requirements Gathering and Analysis, System and Data Modelling, and Testing and Deployment.



STAKEHOLDER INFORMATION NEEDS

Stakeholder	Information Needed	Reason
Project Sponsor / Top Management	Overall project progress, budget usage, milestone completion, risk status	To make high-level decisions and approve phase transitions
Development Team (Marvel, Kennes, Abil, Meini)	Task assignments, technical specifications, deadlines, dependency updates, bug reports	To execute development tasks on schedule and coordinate between modules
Team Leader (Angelo)	Full project status, testing results, deployment readiness, team performance	To lead and control all project phases, especially testing and deployment
End Users / Client	Feature delivery updates, UAT (User Acceptance Testing) schedule, training plan	To prepare for system adoption and provide feedback during testing
QA / Testing Team (Angelo)	Test plans, bug tracking reports, integration test results, deployment checklist	To validate system quality and sign off before go-live

COMMUNICATION PLAN: STAKEHOLDER ANALYSIS

Stakeholder	Document Name	Document Format	Contact Person	Due	Information Content
Phase 1: Project Initiation and Planning					
All Team Members	Project Kick-off Document	Hard copy and Meeting	Angelo (Team Leader)	February 7, 2026	Team roles, project goals, scope, timeline overview
Top Management / Sponsor	Project Scope and Objectives Report	Hard copy	Angelo (Team Leader)	February 7, 2026	Approved project scope, objectives, and initial resource plan
Phase 2: Requirements Gathering and Analysis					
Development Team	System Architecture Document	E-mail and Meeting	Nabila	February 20, 2026	System architecture design, component diagram, tech stack decisions
Development Team	UI/UX Prototype Report	E-mail	Meini	March 7, 2026	Wireframes, user flow diagrams, UI design guidelines
Development Team	Database Design Document	E-mail	Kennes	March 7, 2026	Database schema, entity relationships, data dictionary
All Team Members	Functional Requirements Specification	Hard copy and Meeting	Kennes	March 8, 2026	Full list of system features, functional and non-functional requirements

COMMUNICATION PLAN: STAKEHOLDER ANALYSIS

Stakeholder	Document Name	Document Format	Contact Person	Due	Information Content
Phase 3: System and Data Modelling					
Development Team	Weekly Progress Report	E-mail	Angelo (Team Leader)	Every Monday (Mar to May)	Status of each module, blockers, percentage completion per task
Top Management	Monthly Status Report	Hard copy and Meeting	Angelo (Team Leader)	1st of each month	High-level milestone progress, risks, budget utilization, schedule adherence
Christofer (Core Dev)	Core Module Completion Report	E-mail	Christofer	March 8, 23, 31 and April 22	Auth, POS, Receipt, Product Mgmt, P&L module delivery status
Kennes (Ops Dev)	Operations Module Report	E-mail	Kennes	April 7, 15, and 22	Inventory, Stock Alert, Supplier, Restock module delivery status
Nabila (Finance Dev)	Financial Report Module Update	E-mail	Nabila	April 22 and May 8	Sales and Cashflow report module progress and test results
Meini (Analytics Dev)	Analytics Module Report	E-mail	Meini	May 15	Dashboard Analytics and Export Reports completion status

COMMUNICATION PLAN: STAKEHOLDER ANALYSIS

Stakeholder	Document Name	Document Format	Contact Person	Due	Information Content
Phase 4: Testing and Deployment					
QA / Testing Team	Security and Validation Test Report	Hard copy	Angelo (Team Leader)	May 23, 2026	Security vulnerability results, data validation test cases and outcomes
All Team Members	Integration Test and Bug Fix Report	Hard copy and Meeting	Angelo (Team Leader)	May 31, 2026	Bug list, severity levels, fix status, regression test results
Top Management / Client	Deployment and Documentation Report	Hard copy	Angelo (Team Leader)	May 31, 2026	Go-live confirmation, user manual, system documentation, handover checklist

COMMUNICATION METHODS AND GUIDELINES

COMMUNICATION METHODS



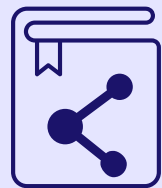
Hard Copy & Meeting

Used for high-stakes communications with management and the full team; ensures formal acknowledgment



Email

Used for technical updates between developers; provides a written trail; efficient for async communication



Shared Drive

Used for internal documentation accessible to all team members at any time

COMMUNICATION FREQUENCY



Weekly

Progress reports every Monday during Phase 3 (Mar to May)



Monthly

Formal status reports on the 1st of each month to management



Phase-Based

Key deliverable documents at each phase milestone

ESCALATION PATH

Weekly Reports → Unresolved Issues → Angelo (24h) → Monthly Status Report / Emergency Briefing → Top Management

COMMUNICATION PLAN: PROJECT TEAM CONTACT DIRECTORY

Name	Role	Phase Responsibility	Documents Owned	Format
Angelo	Team Leader / QA	Phase 1 and Phase 4 (Testing and Deployment)	Kick-off Doc, Scope Report, Test Reports, Deployment Report	Hard copy, Meeting
Abil	Developer: System Architecture	Phase 2 (Architecture)	System Architecture Doc, Financial	E-mail, Meeting
		and Phase 3 (Finance)	Report (Sales), Cashflow Report	
Meini	Developer: UI/UX and Analytics	Phase 2 (UI/UX) and Phase 3 (Analytics)	UI/UX Prototype, Dashboard Analytics, Export Reports	E-mail
<u>Kennes</u>	Developer: Database and Operations	Phase 2 (DB) and Phase 3 (Ops)	DB Design Doc, Functional Req., Inventory, Supplier, Restock Modules	E-mail, Hard copy
Marvel	Developer: Core System	Phase 3 (Core modules)	Auth, POS, Receipt, Product Mgmt, P&L modules	E-mail

5 YEAR GROWTH PLAN SNAPCASH



YEAR 1 • 2026

PRODUCT VALIDATION & MARKET ENTRY

FOCUS

Finalize product development, run pilot tests, and achieve market validation with 50–100 active MSME users in Greater Jakarta (Jabodetabek) through free trials, referrals, and direct outreach to small F&B businesses.

Development Focus

- POS System Refinement
- Inventory Management
- Financial Reporting
- Dashboard Analytics
- Basic Customer Support

100

Target Customers

70%

Retention Rate

10

MSME Case Studies

PRODUCT-MARKET FIT

Milestone

YEAR 2 • 2027

REGIONAL EXPANSION

After market validation, SnapCash expands to major cities: Bandung, Surabaya, and Yogyakarta. Focus shifts from product development to customer acquisition and user experience improvement. A sales and customer success team is built to onboard MSMEs unfamiliar with digital tools.

Development Focus

- Mobile Application
- QRIS Integration
- Multi-user Access
- Cloud Backup System

Target KPIs

300

Active Customers

< 8 %

Churn Rate

90%

Customer
Satisfaction

~

Break-even
Approaching

YEAR 3 • 2028

NATIONAL SCALING

SnapCash evolves from a cashier system into a full business decision-making tool powered by data analytics and basic automation. Partnerships are established with POS hardware distributors, MSME communities, and digital payment providers.

Development Focus

- AI-based Sales Forecasting
- Inventory Prediction
- Loyalty Program
- Multi-Outlet Management

Target KPIs

1000

Active Customers

50

Reseller Partners

90%

Annual Revenue



Stable Profitability